

EA-0014 - AQELYN Engineering Archive

AQELYN - EA-0014 Engineering Archive

IS-014 - AQELYN Threat Intelligence Fusion Engine

****Archive ID:**** EA-0014

****Implementation Specification:**** IS-014

****Component:**** AQELYN Threat Intelligence Fusion Engine

****Project:**** AQELYN

****System Type:**** Cyber Security Operating Environment

****Status:**** COMPLETE

****Repository Impact:**** No top-level repository structure changes

****Breaking Changes:**** None

****Engineering Phase:**** Phase 3

****Predecessor Archives:**** EA-0001 through EA-0013

****Next Specification:**** IS-015

Document Control

| Field | Value |

---|---

| Document | Engineering Archive EA-0014 |

| Specification | IS-014 - AQELYN Threat Intelligence Fusion Engine |

| Publication Format | Markdown, PDF, HTML, ZIP |

| Source of Truth | MD/EA-0014.md |

| Archive Rule | Implementation Specification -> Engineering Archive -> Continue |

| Repository Rule | Fixed repository structure; no redesign |

| Completion State | IS-014 complete; EA-0014 generated |

1. Engineering Context

AQELYN is being built as a modular Cyber Security Operating Environment.

Completed engineering archive chain:

| Engineering Archive | Implementation Specification | Component |

---|---

| EA-0001 | IS-001 | AQELYN Kernel |

| EA-0002 | IS-002 | Universal Object Model |

| EA-0003 | IS-003 | AQELYN Event Bus |

EA-0004	IS-004	AQELYN Evidence Engine
EA-0005	IS-005	AQELYN Knowledge Graph
EA-0006	IS-006	AQELYN Trust Engine
EA-0007	IS-007	AQELYN Mission Engine
EA-0008	IS-008	AQELYN Workflow Engine
EA-0009	IS-009	AQELYN Policy Engine
EA-0010	IS-010	AQELYN Compliance & Governance Engine
EA-0011	IS-011	AQELYN Identity & Access Governance Engine
EA-0012	IS-012	AQELYN Asset & Configuration Governance Engine
EA-0013	IS-013	AQELYN Risk Intelligence Engine
EA-0014	IS-014	AQELYN Threat Intelligence Fusion Engine

The fixed repository structure remains:

```
AQELYN/  
■■■ archive/  
■■■ blueprint/  
■■■ docs/  
■■■ src/  
■■■ tests/  
■■■ tools/  
■■■ build/  
■■■ releases/  
■■■ scripts/  
■■■ assets/  
■■■ examples/  
■■■ plugins/  
■■■ sdk/  
■■■ api/  
■■■ README.md
```

The engineering rule remains:

```
Finish Implementation Specification  
↓  
Generate Engineering Archive  
↓  
Continue
```

No Engineering Archive may be skipped.

2. IS-014 Specification Identity

```
Specification ID: IS-014  
Name: AQELYN Threat Intelligence Fusion Engine  
Engineering Archive Target: EA-0014  
Project: AQELYN  
System Type: Cyber Security Operating Environment  
Status: Complete  
Predecessor: IS-013 - AQELYN Risk Intelligence Engine
```

3. Purpose

The AQELYN Threat Intelligence Fusion Engine ingests, normalizes, correlates, scores, and operationalizes threat intelligence from internal and external sources.

It transforms disconnected threat feeds into actionable intelligence for risk, mission, asset, identity, policy, and workflow decisions.

It answers:

What threats are relevant to us?
Which indicators are trustworthy?
Which adversaries target our assets or missions?
Which risks are increasing because of threat activity?
Which evidence supports the intelligence?
Which workflows should be triggered?

4. Mission

The engine shall provide:

Threat intelligence ingestion
Indicator normalization
Threat source trust scoring
Threat correlation
Adversary profiling
Campaign tracking
TTP mapping
Indicator lifecycle management
Threat-to-risk mapping
Threat-to-asset mapping
Threat-to-mission mapping
Evidence-backed intelligence
Operational threat reporting

5. Scope

5.1 In Scope

Threat feed ingestion
Indicator catalog
Threat actor catalog
Campaign catalog
TTP mapping
Source confidence scoring
Indicator confidence scoring
Threat correlation
Threat enrichment
Risk intelligence integration
Mission impact integration
Evidence binding
Threat event publishing

5.2 Out of Scope

Full malware sandbox implementation
Endpoint detection engine
Network packet inspection engine
Dark web marketplace monitoring
Human intelligence operations
Law enforcement case management
Offensive cyber operations

6. Dependencies

IS-014 depends on:

IS-001 AQELYN Kernel
IS-002 Universal Object Model
IS-003 AQELYN Event Bus
IS-004 AQELYN Evidence Engine
IS-005 AQELYN Knowledge Graph
IS-006 AQELYN Trust Engine
IS-007 AQELYN Mission Engine

```
IS-008 AQELYN Workflow Engine
IS-009 AQELYN Policy Engine
IS-010 AQELYN Compliance & Governance Engine
IS-011 AQELYN Identity & Access Governance Engine
IS-012 AQELYN Asset & Configuration Governance Engine
IS-013 AQELYN Risk Intelligence Engine
```

7. High-Level Architecture

```
AQELYN Threat Intelligence Fusion Engine
■
■■■ Threat Source Registry
■■■ Threat Feed Ingestion Service
■■■ Indicator Normalization Service
■■■ Indicator Catalog
■■■ Threat Actor Catalog
■■■ Campaign Tracker
■■■ TTP Mapping Service
■■■ Threat Correlation Engine
■■■ Threat Confidence Scoring Engine
■■■ Threat Evidence Binder
■■■ Risk Intelligence Connector
■■■ Mission Impact Connector
■■■ Threat Reporting Service
■■■ Threat Event Publisher
```

8. Functional Requirements

FR-014-001 - Threat Source Registry

The system shall maintain a registry of threat intelligence sources.

Each source shall include:

```
source_id
name
source_type
authority
trust_score
confidence_model
update_frequency
license_scope
status
```

FR-014-002 - Threat Feed Ingestion

The system shall ingest threat intelligence from:

```
Commercial feeds
Open-source feeds
Government advisories
Internal detections
Incident reports
Security vendor reports
ISAC/ISAO feeds
Manual analyst submissions
```

FR-014-003 - Indicator Catalog

The system shall maintain normalized indicators, including:

```
IP addresses
Domains
URLs
```

- File hashes
- Email addresses
- Certificates
- Registry keys
- User agents
- Malware names
- Infrastructure identifiers

FR-014-004 - Threat Actor Catalog

The system shall maintain profiles for threat actors and adversary groups.

FR-014-005 - Campaign Tracking

The system shall track threat campaigns and associate them with:

- Actors
- Indicators
- TTPs
- Targets
- Affected sectors
- Observed timeframes
- Evidence

FR-014-006 - TTP Mapping

The system shall map observed tactics, techniques, and procedures to normalized technique references.

FR-014-007 - Confidence Scoring

The system shall calculate confidence for:

- Threat source
- Indicator
- Actor attribution
- Campaign association
- TTP mapping
- Threat relevance

FR-014-008 - Threat Correlation

The system shall correlate threat intelligence with:

- Assets
- Configurations
- Identities
- Missions
- Policies
- Compliance findings
- Risks
- Evidence

FR-014-009 - Risk Integration

The engine shall provide threat context to IS-013 Risk Intelligence.

FR-014-010 - Threat Reporting

The engine shall produce threat intelligence reports for:

- Executives
- Security analysts
- Risk owners
- Mission owners

Compliance stakeholders
Engineering teams

9. Non-Functional Requirements

The engine shall satisfy:

Traceability
Source confidence
Evidence-backed intelligence
Auditability
Timeliness
De-duplication
Explainability
Scalability
Event-driven synchronization
Repository stability
Backward compatibility

10. Core Threat Intelligence Flow

Threat source registered
↓
Feed ingested
↓
Indicators normalized
↓
Source confidence applied
↓
Threat correlated
↓
Evidence linked
↓
Risk and mission relevance calculated
↓
Threat report generated
↓
Threat event published

11. Internal Component Architecture

The AQELYN Threat Intelligence Fusion Engine is implemented as a modular, event-driven subsystem integrated with the AQELYN Event Bus.

AQELYN Threat Intelligence Fusion Engine

-
- Threat Source Registry
- Feed Ingestion Service
- Feed Normalization Service
- Indicator Catalog
- Threat Actor Registry
- Campaign Manager
- TTP Mapping Service
- Threat Correlation Engine
- Confidence Scoring Engine
- Evidence Binding Service
- Threat Analytics Engine
- Threat Reporting Service
- Risk Intelligence Connector
- Mission Intelligence Connector
- Threat Event Publisher

Each service shall operate independently while sharing the Universal Object Model and Evidence Engine.

12. Component Specifications

12.1 Threat Source Registry

Maintains metadata for every intelligence provider.

Responsibilities:

- Source registration
- Source classification
- Trust scoring
- Licensing metadata
- Feed health monitoring
- Update scheduling
- Source lifecycle

12.2 Feed Ingestion Service

Responsible for importing intelligence.

Supported formats:

- STIX
- TAXII
- JSON
- XML
- CSV
- REST APIs
- Streaming feeds
- Manual analyst uploads

The ingestion layer shall support incremental synchronization and deduplication.

12.3 Feed Normalization Service

Converts heterogeneous threat intelligence into the AQELYN Universal Object Model.

Normalization includes:

- Indicator normalization
- Timestamp normalization
- Identity normalization
- Threat actor normalization
- Campaign normalization
- TTP normalization
- Confidence normalization

12.4 Indicator Catalog

Stores normalized Indicators of Compromise.

Supported indicator classes:

- IPv4
- IPv6
- Domain
- FQDN
- URL
- URI
- Email
- SHA256
- SHA1
- MD5
- X509 Certificate
- Mutex
- Registry Key

```
User Agent
File Name
ASN
```

12.5 Threat Actor Registry

Maintains profiles for adversaries.

Attributes include:

```
Actor ID
Aliases
Country
Motivation
Capabilities
Known campaigns
Known TTPs
Known infrastructure
Associated evidence
Confidence
```

12.6 Campaign Manager

Maintains campaign intelligence.

Each campaign contains:

```
Campaign ID
Name
Actors
Objectives
Timeframe
Affected sectors
Indicators
Evidence
Status
```

12.7 TTP Mapping Service

Maps observed behavior to standardized tactics and techniques.

Supported mappings include:

```
Tactics
Techniques
Sub-techniques
Detection guidance
Mitigation guidance
Evidence references
```

12.8 Threat Correlation Engine

Correlates intelligence across AQELYN.

Correlation targets:

```
Assets
Configurations
Identities
Missions
Policies
Compliance findings
Risk records
Knowledge Graph
Evidence
```

12.9 Confidence Scoring Engine

Calculates confidence for:

- Threat source
- Indicator validity
- Threat actor attribution
- Campaign attribution
- TTP mapping
- Correlation quality
- Overall intelligence quality

Confidence shall be recalculated whenever new evidence becomes available.

12.10 Evidence Binding Service

Every intelligence object shall reference immutable evidence.

Evidence sources include:

- Threat feeds
- Internal detections
- Incident investigations
- SOC observations
- Analyst submissions
- External advisories

12.11 Threat Analytics Engine

Provides analytical capabilities including:

- Trend analysis
- Campaign evolution
- Indicator reuse
- Threat actor activity
- Sector targeting
- Infrastructure reuse
- Emerging threat detection

12.12 Threat Reporting Service

Produces operational and executive reports.

Reports include:

- Threat landscape
- Top campaigns
- Top actors
- Emerging threats
- Threat trends
- Indicator statistics
- Threat confidence
- Risk impact
- Mission impact

13. Universal Object Model Extensions

13.1 ThreatSource

- ThreatSource:
- source_id
- name
- type
- trust_score
- confidence

```
update_frequency
```

13.2 ThreatIndicator

```
ThreatIndicator:  
indicator_id  
indicator_type  
value  
confidence  
first_seen  
last_seen  
source
```

13.3 ThreatActor

```
ThreatActor:  
actor_id  
aliases  
motivation  
capabilities  
confidence
```

13.4 ThreatCampaign

```
ThreatCampaign:  
campaign_id  
actors  
indicators  
ttps  
timeframe
```

13.5 ThreatAssessment

```
ThreatAssessment:  
assessment_id  
object_id  
confidence  
evidence
```

14. Knowledge Graph Integration

Relationships include:

```
Threat Actor  
↓  
conducts  
↓  
Campaign  
  
Campaign  
↓  
uses  
↓  
Indicators  
  
Indicators  
↓  
target  
↓  
Assets  
  
Indicators  
↓  
target  
↓  
Identities  
  
Threat  
↓  
increases
```

```

↓
Risk
Threat
↓
affects
↓
Mission
Threat
↓
supported_by
↓
Evidence

```

15. Event Bus Integration

The engine publishes standardized events.

15.1 Threat Events

```

threat.source.created
threat.source.updated
indicator.created
indicator.updated
campaign.created
campaign.updated
actor.created
actor.updated

```

15.2 Correlation Events

```

threat.correlated
indicator.matched
campaign.detected
actor.correlated

```

15.3 Intelligence Events

```

threat.confidence.changed
threat.assessed
threat.enriched
threat.archived

```

15.4 Risk Events

```

risk.threat.updated
mission.threat.updated

```

16. Evidence Engine Integration

The Threat Intelligence Fusion Engine consumes immutable evidence from:

```

External threat feeds
Internal detections
SOC investigations
Compliance findings
Asset observations
Identity observations
Incident response

```

Evidence references shall remain immutable after publication.

17. Policy Engine Integration

Policies determine:

- Accepted threat sources
- Confidence thresholds
- Indicator retention
- Threat retention
- Campaign retention
- Alert thresholds
- Publishing rules

18. Risk Intelligence Integration

IS-013 consumes threat context from this engine.

Shared information includes:

- Threat confidence
- Campaign relevance
- Actor attribution
- Threat indicators
- Risk impact
- Evidence confidence

19. Mission Engine Integration

Mission Engine consumes:

- Mission targeting
- Mission disruption probability
- Threat priority
- Operational relevance

Mission priorities influence threat prioritization.

20. Asset Governance Integration

Threat intelligence correlates with:

- Critical assets
- Configuration drift
- Asset exposure
- Asset ownership

21. Identity Governance Integration

Threat intelligence correlates with:

- Privileged identities
- Compromised identities
- Credential exposure
- Identity anomalies

22. Compliance Integration

Threat intelligence provides context for:

- Control effectiveness
- Regulatory exposure
- Security posture
- Compliance exceptions

23. Public APIs

23.1 Threat Source API

```
GET /threat-sources
POST /threat-sources
GET /threat-sources/{id}
```

23.2 Indicator API

```
GET /indicators
POST /indicators
GET /indicators/{id}
PUT /indicators/{id}
```

23.3 Threat Actor API

```
GET /actors
POST /actors
GET /actors/{id}
```

23.4 Campaign API

```
GET /campaigns
POST /campaigns
GET /campaigns/{id}
```

23.5 Analytics API

```
GET /threat-analytics
GET /threat-trends
GET /threat-dashboard
```

23.6 Correlation API

```
POST /threat/correlate
POST /indicator/match
POST /campaign/analyze
```

24. Repository Impact

Implementation shall follow the approved AQELYN repository without modification.

```
AQELYN/
├── src/
│   ├── threat_intelligence/
│   ├── tests/
│   ├── threat_intelligence/
│   ├── docs/
│   ├── threat_intelligence/
│   ├── api/
│   ├── threat_intelligence/
│   └── archive/
```

No top-level repository changes are permitted.

25. Security Architecture

The AQELYN Threat Intelligence Fusion Engine is a Tier-1 intelligence subsystem responsible for ingesting and operationalizing external and internal threat data.

Every threat intelligence decision shall be:

- Evidence-backed
- Source-attributed
- Confidence-scored
- Policy-governed
- Auditable
- Traceable
- Explainable
- Continuously re-evaluated

No threat intelligence object shall be operationalized without a source, confidence state, and evidence reference.

25.1 Security Principles

The engine shall implement:

- Zero Trust
- Source Verification
- Evidence Integrity
- Least Privilege
- Defense in Depth
- De-duplication Integrity
- Immutable Intelligence History
- Explainable Correlation
- Separation of Duties
- Security by Design

25.2 Threat Intelligence Authorization Model

Only authorized roles may register sources, approve intelligence sources, change confidence rules, or operationalize high-impact intelligence.

Supported roles:

Role	Responsibility
Threat Intelligence Administrator	Source and feed lifecycle management
Threat Analyst	Indicator, campaign, actor, and TTP assessment
SOC Analyst	Operational intelligence review
Security Officer	High-impact intelligence approval
Mission Owner	Mission relevance validation
Risk Owner	Risk impact review
Automation Service	Approved ingestion, scoring, and correlation

Authorization decisions shall be enforced through the AQELYN Policy Engine.

25.3 Source Integrity

Threat sources shall maintain:

```
Source Identifier
Authority
Trust Score
License Scope
Update Frequency
Health State
Observed Reliability
Evidence References
Review History
```

Source trust scores shall be recalculated when quality, reliability, or attribution changes.

25.4 Indicator Integrity

Threat indicators shall maintain:

```
Indicator Identifier
Indicator Type
Normalized Value
Original Value
Source
Confidence
First Seen
Last Seen
Evidence References
Lifecycle State
```

Indicators shall not be modified destructively. Corrections and updates shall be recorded as new versions.

25.5 Intelligence Evidence Protection

Threat intelligence evidence shall support:

```
Immutable evidence references
Evidence lineage
Source attribution
Version tracking
Integrity verification
Trust scoring
Audit history
```

The engine shall not overwrite source evidence.

26. Threat Intelligence Lifecycle

Every governed threat intelligence object follows a controlled lifecycle.

```
Received
↓
Normalized
↓
Validated
↓
Scored
↓
Correlated
↓
Operationalized
↓
Monitored
↓
Expired
↓
Archived
```

26.1 Indicator Lifecycle

```
Observed
↓
Normalized
↓
Deduplicated
↓
Confidence Scored
↓
Correlated
↓
Published
↓
Expired
↓
Archived
```

26.2 Threat Actor Lifecycle

```
Identified
↓
Profiled
↓
Attributed
↓
Correlated
↓
Reviewed
↓
Updated
↓
Archived
```

26.3 Campaign Lifecycle

```
Detected
↓
Scoped
↓
Associated
↓
Tracked
↓
Assessed
↓
Closed
↓
Archived
```

26.4 Source Lifecycle

```
Proposed
↓
Approved
↓
Active
↓
Monitored
↓
Suspended
↓
Retired
```

27. Threat Evaluation Model

Threat intelligence evaluation shall consider:

```
Source Trust
Indicator Confidence
Actor Attribution Confidence
Campaign Confidence
TTP Mapping Confidence
Evidence Quality
```

- Internal Correlation
- External Corroboration
- Operational Relevance
- Mission Relevance
- Risk Impact

27.1 Threat Confidence Levels

- Unknown
- Low
- Moderate
- High
- Confirmed

27.2 Threat Relevance Levels

- Not Relevant
- Contextual
- Relevant
- High Priority
- Mission Critical

Mission-critical threat intelligence shall trigger immediate governance workflows where configured.

28. Continuous Threat Intelligence Fusion

The engine shall continuously monitor:

- New indicators
- Updated indicators
- Actor attribution changes
- Campaign evolution
- Source reliability changes
- Threat feed health
- Asset exposure matches
- Identity exposure matches
- Risk score changes
- Mission relevance changes
- Compliance implications
- Evidence updates

Every significant intelligence change shall trigger reassessment and appropriate events.

29. Audit & Reporting

29.1 Executive Reports

- Threat Landscape Overview
- Top Threats
- Mission-Relevant Threats
- Threat-Driven Risk Summary
- Threat Trends

29.2 Operational Reports

- Active Indicators
- Campaign Watchlist
- Threat Actor Activity
- Indicator Match Results
- Source Health

29.3 Risk Reports

Threat-to-Risk Mapping
Risk Impact of Campaigns
Threat-Increased Risk
Threat Confidence Summary

29.4 Engineering Reports

Feed Processing Statistics
Correlation Statistics
Indicator Deduplication Metrics
Evidence Coverage
Confidence Model Metrics

30. Failure Handling

30.1 Source Unavailable

Status:
Feed Degraded

Action:
Continue using cached intelligence and mark source health degraded

30.2 Malformed Feed Data

Status:
Ingestion Failed

Action:
Reject malformed objects and record ingestion error evidence

30.3 Missing Evidence

Status:
Intelligence Incomplete

Action:
Prevent operationalization until evidence is linked

30.4 Confidence Calculation Failure

Status:
Assessment Pending

Action:
Queue for recalculation and analyst review

30.5 Correlation Failure

Status:
Correlation Pending

Action:
Retry according to correlation policy

30.6 Event Bus Interruption

Threat intelligence events shall be queued until successful delivery.

31. Performance Requirements

The engine shall support:

High-volume feed ingestion
Incremental feed synchronization
Indicator deduplication at scale
Parallel correlation
Asynchronous enrichment
Low-latency event publishing
Dashboard generation

Threat ingestion shall not block operational AQELYN services.

32. Scalability Requirements

The architecture shall support:

Millions of indicators
Hundreds of threat feeds
Large campaign histories
Large actor catalogs
Multi-tenant deployments
Hybrid environments
Distributed processing
High-volume correlation workloads

33. Testing Strategy

33.1 Unit Testing

Validate:

Threat Source Registry
Feed Ingestion Service
Feed Normalization Service
Indicator Catalog
Threat Actor Registry
Campaign Manager
TTP Mapping Service
Confidence Scoring Engine
Threat Correlation Engine
Evidence Binding Service

33.2 Integration Testing

Verify interaction with:

AQELYN Kernel
Universal Object Model
Event Bus
Evidence Engine
Knowledge Graph
Trust Engine
Mission Engine
Workflow Engine
Policy Engine
Compliance Engine
Identity Governance Engine
Asset Governance Engine
Risk Intelligence Engine

33.3 System Testing

Validate end-to-end scenarios including:

Threat source registration
Feed ingestion

```
Indicator normalization
Indicator deduplication
Confidence scoring
Threat correlation
Risk update
Mission threat update
Threat report generation
```

33.4 Security Testing

Verify:

```
Authorization
Source integrity
Indicator integrity
Evidence protection
Audit trail completeness
Policy enforcement
Workflow security
```

33.5 Regression Testing

Ensure IS-001 through IS-013 continue operating without behavioral changes introduced by IS-014.

34. Acceptance Criteria

IS-014 shall be considered complete when:

```
Threat Source Registry is defined.
Feed Ingestion Service is documented.
Indicator Catalog is defined.
Threat Actor Registry is defined.
Campaign Manager is documented.
TTP Mapping Service is defined.
Confidence Scoring Engine is documented.
Threat Correlation Engine is defined.
Evidence Binding Service is documented.
Integration with IS-001 through IS-013 is complete.
Repository structure remains unchanged.
Testing strategy is documented.
```

35. Repository Validation

Implementation shall follow the approved repository structure.

```
AQELYN/
■■■ src/threat_intelligence/
■■■ tests/threat_intelligence/
■■■ docs/threat_intelligence/
■■■ api/threat_intelligence/
■■■ archive/
```

No repository redesign is introduced.

36. Engineering Summary

The AQELYN Threat Intelligence Fusion Engine provides the intelligence layer that transforms external and internal threat data into evidence-backed operational threat context.

It introduces:

- Threat Source Registry
- Feed Ingestion
- Feed Normalization
- Indicator Catalog
- Threat Actor Registry
- Campaign Management
- TTP Mapping
- Confidence Scoring
- Threat Correlation
- Evidence Binding
- Threat Analytics
- Threat Reporting
- Risk and Mission Integration

The engine integrates with all previous AQELYN components while preserving modularity and backward compatibility.

37. Specification Status

Specification ID : IS-014
Title : AQELYN Threat Intelligence Fusion Engine
Status : COMPLETE
Engineering Archive : READY FOR GENERATION
Next Artifact : EA-0014

Engineering workflow status:

- EA-0001 COMPLETE
- EA-0002 COMPLETE
- EA-0003 COMPLETE
- EA-0004 COMPLETE
- EA-0005 COMPLETE
- EA-0006 COMPLETE
- EA-0007 COMPLETE
- EA-0008 COMPLETE
- EA-0009 COMPLETE
- EA-0010 COMPLETE
- EA-0011 COMPLETE
- EA-0012 COMPLETE
- EA-0013 COMPLETE
- IS-014 COMPLETE
- EA-0014 READY FOR GENERATION

38. EA-0014 Engineering Objective

The objective of IS-014 was to introduce a dedicated Threat Intelligence Fusion Engine that enables AQELYN to ingest, normalize, correlate, score, and operationalize threat intelligence from trusted internal and external sources.

The engine extends AQELYN governance and risk intelligence with actionable threat context for mission, asset, identity, policy, compliance, workflow, and executive decisions.

39. EA-0014 Engineering Summary

The implementation specification defines a modular subsystem responsible for:

- Threat source registration
- Threat feed ingestion
- Feed normalization
- Indicator cataloging
- Threat actor registry
- Campaign tracking
- TTP mapping
- Threat correlation

```
Confidence scoring
Evidence binding
Threat analytics
Threat reporting
Risk intelligence integration
Mission intelligence integration
Threat event publishing
```

The engine integrates with all previously completed AQELYN engines while preserving architectural modularity.

40. Major Engineering Decisions

40.1 Decision 1 - Dedicated Threat Intelligence Fusion Engine

Threat intelligence responsibilities are implemented as a standalone engine rather than embedded in Risk Intelligence, Asset Governance, or Mission.

Rationale:

```
Clear separation of intelligence ingestion from risk analytics.
Independent lifecycle and scaling.
Better support for threat feed normalization and deduplication.
Improved traceability of source confidence and indicator lifecycle.
```

40.2 Decision 2 - Source-Attributed Intelligence

Every intelligence object must maintain a source and evidence reference.

Benefits:

```
Threat intelligence becomes auditable.
Indicator confidence is explainable.
Operational decisions can be traced to source material.
Risk and mission decisions can consume trusted threat context.
```

40.3 Decision 3 - Confidence-Driven Operationalization

Threat intelligence shall be operationalized based on confidence, source trust, evidence quality, and policy thresholds.

Benefits:

```
Low-quality intelligence does not automatically trigger action.
Analyst review can be required for ambiguous intelligence.
Policy controls can govern retention, publishing, and escalation.
```

40.4 Decision 4 - Event-Driven Threat Fusion

Threat, indicator, actor, campaign, correlation, confidence, risk, and mission updates are published through the AQELYN Event Bus.

Examples include:

```
threat.source.created
indicator.created
campaign.updated
actor.updated
threat.correlated
indicator.matched
threat.confidence.changed
risk.threat.updated
mission.threat.updated
```

This maintains loose coupling between AQELYN engines.

40.5 Decision 5 - Universal Object Model Extension

New domain objects introduced include:

```
ThreatSource
ThreatIndicator
ThreatActor
ThreatCampaign
ThreatAssessment
```

These extend the Universal Object Model without modifying existing object definitions.

41. Architectural Integration Summary

| Engine | Integration |

|---|---|

| IS-001 Kernel | Runtime lifecycle and service registration |

| IS-002 Universal Object Model | Threat source, indicator, actor, campaign, assessment objects |

| IS-003 Event Bus | Threat, indicator, correlation, confidence, risk, mission events |

| IS-004 Evidence Engine | Immutable feed, advisory, analyst, investigation, and detection evidence |

| IS-005 Knowledge Graph | Actor, campaign, indicator, asset, identity, risk, mission relationships |

| IS-006 Trust Engine | Source trust, indicator confidence, evidence confidence |

| IS-007 Mission Engine | Mission targeting and mission threat relevance |

| IS-008 Workflow Engine | Analyst review, escalation, enrichment, operationalization workflows |

| IS-009 Policy Engine | Source acceptance, confidence thresholds, retention and publishing rules |

| IS-010 Compliance Engine | Control effectiveness, regulatory exposure, compliance exceptions |

| IS-011 Identity Governance Engine | Compromised identities, credential exposure, privileged identity context |

| IS-012 Asset Governance Engine | Critical assets, configuration drift, exposure, ownership |

| IS-013 Risk Intelligence Engine | Threat-driven risk scoring and risk updates |

No existing engine required redesign.

42. Repository Impact Summary

Repository structure remains unchanged.

Implementation is expected within existing project directories, including:

```
AQELYN/
■■■ src/threat_intelligence/
■■■ tests/threat_intelligence/
■■■ api/threat_intelligence/
■■■ docs/threat_intelligence/
■■■ archive/
```

No top-level directories were added, removed, or renamed.

43. Security Impact Summary

The specification introduces threat-intelligence-specific security controls:

- Policy-driven threat source authorization
- Source integrity
- Indicator integrity
- Evidence-backed intelligence records
- Immutable intelligence history
- Confidence-based operationalization
- Separation of duties for high-impact intelligence
- Traceable analyst and automation decisions

No reduction in the security posture of existing components was identified.

44. Capabilities Added

The engine enables AQELYN to support:

- Threat source registry
- Threat feed ingestion
- Feed normalization
- Indicator catalog
- Threat actor registry
- Campaign tracking
- TTP mapping
- Confidence scoring
- Threat correlation
- Threat analytics
- Threat reporting
- Risk intelligence integration
- Mission threat integration
- Asset threat context
- Identity threat context
- Compliance threat context
- Evidence-backed intelligence

45. Risks Identified

| Risk | Mitigation |

|---|---|

Low-quality threat feeds	Source trust scoring and policy thresholds
Malformed feed data	Ingestion validation and rejection with evidence
False positive indicators	Confidence scoring, deduplication, and correlation rules
Over-attribution of threat actors	Attribution confidence and analyst review
Excessive feed volume	Incremental ingestion, deduplication, and parallel processing
License misuse	Source license metadata and policy enforcement
Stale indicators	Indicator lifecycle and expiration
Missing evidence	Prevent operationalization until evidence is linked

No critical architectural risks were identified that require redesign.

46. Verification Summary

The specification defines verification for:

- Unit testing
- Integration testing
- System testing
- Security testing

Regression testing

Acceptance criteria cover source registry, feed ingestion, indicator catalog, actor registry, campaign management, TTP mapping, confidence scoring, threat correlation, evidence binding, integration with IS-001 through IS-013, repository validation, and testing documentation.

47. Engineering Principles Confirmed

The implementation complies with established AQELYN principles:

Modular architecture
Event-driven communication
Immutable evidence references
Traceability
Explainability
Security by design
Repository stability
Backward compatibility
Governance-before-archive discipline

48. Dependencies

Required:

EA-0001 through EA-0013
IS-001 through IS-013

Enables:

IS-015 and subsequent threat-dependent components

49. Completion Record

Engineering Archive : EA-0014
Implementation Specification : IS-014
Title : AQELYN Threat Intelligence Fusion Engine
Engineering Status : COMPLETE
Repository Status : UNCHANGED
Architecture Status : EXTENDED
Backward Compatibility : MAINTAINED
Engineering Rule :
IS Completed
↓
EA Generated
↓
Continue

50. Archive Index Update

EA-0001 IS-001 AQELYN Kernel
EA-0002 IS-002 Universal Object Model
EA-0003 IS-003 AQELYN Event Bus
EA-0004 IS-004 AQELYN Evidence Engine
EA-0005 IS-005 AQELYN Knowledge Graph
EA-0006 IS-006 AQELYN Trust Engine
EA-0007 IS-007 AQELYN Mission Engine
EA-0008 IS-008 AQELYN Workflow Engine
EA-0009 IS-009 AQELYN Policy Engine
EA-0010 IS-010 AQELYN Compliance & Governance Engine
EA-0011 IS-011 AQELYN Identity & Access Governance Engine

EA-0012 IS-012 AQELYN Asset & Configuration Governance Engine
 EA-0013 IS-013 AQELYN Risk Intelligence Engine
 EA-0014 IS-014 AQELYN Threat Intelligence Fusion Engine

51. Engineering Phase Status

Completed Engineering Archives : EA-0001 through EA-0014

Current Status:
 EA-0014 COMPLETE

Next Implementation Specification:
 IS-015

EA-0014 is completed and archived. The engineering workflow is consistent with the project rule:

Implementation Specification -> Engineering Archive -> Continue

From this point onward, the next engineering artifact is IS-015.

52. Engineering Archive Publication Standard

EA-0014 follows the AQELYN Engineering Archive Publication Standard.

Required package structure:

```
EA-xxxx/
■
■ diagrams/
■ ■ ■ ■ Architecture.svg
■ ■ ■ ■ Component.svg
■ ■ ■ ■ Workflow.svg
■ ■ ■ ■ EventFlow.svg
■ ■ ■ ■ Integration.svg
■
■ ■ ■ ■ examples/
■ ■ ■ ■ example_*.md
■
■ ■ ■ ■ HTML/
■ ■ ■ ■ index.html
■ ■ ■ ■ styles.css
■ ■ ■ ■ assets/
■
■ ■ ■ ■ images/
■
■ ■ ■ ■ journal/
■ ■ ■ ■ Engineering_Journal.md
■
■ ■ ■ ■ manifest/
■ ■ ■ ■ manifest.json
■
■ ■ ■ ■ MD/
■ ■ ■ ■ EA-xxxx.md
■
■ ■ ■ ■ PDF/
■ ■ ■ ■ EA-xxxx.pdf
■
■ ■ ■ ■ requirements/
■ ■ ■ ■ Requirements_Matrix.md
■
■ ■ ■ ■ traceability/
■ ■ ■ ■ Traceability_Matrix.md
■
■ ■ ■ ■ README.md
```

The master Markdown is the source of truth. The PDF and HTML are generated from the same master Markdown and must not omit sections.

53. Requirements Matrix

Requirement ID	Requirement	Evidence in Archive	Status
---	---	---	---
FR-014-001	Maintain threat source registry	Sections 8, 12	Complete
FR-014-002	Ingest threat feeds	Sections 8, 12	Complete
FR-014-003	Maintain indicator catalog	Sections 8, 12, 13	Complete
FR-014-004	Maintain threat actor catalog	Sections 8, 12, 13	Complete
FR-014-005	Track campaigns	Sections 8, 12, 13	Complete
FR-014-006	Map TTPs	Sections 8, 12	Complete
FR-014-007	Calculate confidence	Sections 8, 12, 27	Complete
FR-014-008	Correlate threat intelligence	Sections 8, 12, 14	Complete
FR-014-009	Integrate with Risk Intelligence	Sections 8, 18	Complete
FR-014-010	Produce threat reports	Sections 8, 12, 29	Complete
NFR-014-001	Traceability	Sections 9, 14, 16, 49	Complete
NFR-014-002	Source confidence	Sections 9, 25, 27	Complete
NFR-014-003	Evidence-backed intelligence	Sections 9, 16, 40	Complete
NFR-014-004	Auditability	Sections 9, 25, 30	Complete
NFR-014-005	Timeliness and scalability	Sections 9, 31, 32	Complete
NFR-014-006	Repository stability	Sections 24, 35, 42	Complete

54. Traceability Matrix

Source	Target	Relationship
---	---	---
IS-014 Purpose	EA-0014 Objective	Defines why the engine exists
Threat Source Registry	FR-014-001	Implements source lifecycle
Feed Ingestion Service	FR-014-002	Implements feed import
Indicator Catalog	FR-014-003	Implements normalized indicators
Threat Actor Registry	FR-014-004	Implements adversary profiles
Campaign Manager	FR-014-005	Implements campaign tracking
TTP Mapping Service	FR-014-006	Implements TTP mapping
Confidence Scoring Engine	FR-014-007	Implements confidence model
Threat Correlation Engine	FR-014-008	Correlates threat context
Risk Intelligence Connector	FR-014-009	Supplies IS-013 threat context
Threat Reporting Service	FR-014-010	Produces threat reports
Evidence Engine Integration	Evidence-backed intelligence	References immutable evidence
Policy Engine Integration	Publishing and confidence rules	Determines source acceptance and thresholds
Mission Engine Integration	Mission threat relevance	Supplies mission impact context
Asset Governance Integration	Asset threat context	Supplies asset exposure and criticality
Identity Governance Integration	Identity threat context	Supplies privileged and compromised identity context
Event Bus Integration	NFR-014 event-driven sync	Publishes threat events
Repository Validation	Repository Standard	Confirms no top-level redesign

55. Engineering Journal

Journal Entry - EA-0014

EA-0014 was created to archive completion of IS-014 - AQELYN Threat Intelligence Fusion Engine.

The archive records the expansion of AQELYN into threat intelligence fusion. IS-014 defines the structure needed to register sources, ingest feeds, normalize indicators, maintain actor and campaign intelligence, map TTPs, score confidence, correlate intelligence, bind evidence, update risk, update mission threat context, and generate threat reports.

The engineering design preserves the fixed AQELYN repository structure and maintains backward compatibility with previously completed engines.

Lessons Learned

Threat intelligence must be modeled separately from risk intelligence. Risk intelligence consumes threat context, but the Threat Intelligence Fusion Engine owns feed ingestion, source confidence, indicator lifecycle, actor attribution, campaign tracking, and threat correlation.

Governance Note

EA-0014 follows the master-document publication workflow. The Markdown file is the authoritative source, and PDF/HTML representations are generated from the same content.

56. Examples

56.1 Example Threat Source

```
source_id: THSRC-0001
name: Internal SOC Detections
type: internal
trust_score: 0.94
confidence: high
update_frequency: continuous
status: active
```

56.2 Example Threat Indicator

```
indicator_id: IOC-1001
indicator_type: domain
value: malicious-example.test
confidence: high
first_seen: 2026-07-07T12:00:00Z
last_seen: 2026-07-07T12:30:00Z
source: THSRC-0001
```

56.3 Example Threat Campaign

```
campaign_id: CAMP-2001
actors:
- ACTOR-3001
indicators:
- IOC-1001
ttps:
- TTP-PHISHING
timeframe:
first_seen: 2026-07-01
last_seen: 2026-07-07
status: active
```

56.4 Example Threat Event

```
{
  "event_type": "risk.threat.updated",
  "risk_id": "RISK-0001",
  "indicator_id": "IOC-1001",
  "confidence": "high",
  "reason": "Indicator matched mission-critical asset exposure",
  "source_engine": "aqelyn_threat_intelligence_fusion_engine"
}
```

57. Manifest Summary

Archive contents include:

```
README.md
MD/EA-0014.md
PDF/EA-0014.pdf
HTML/index.html
HTML/styles.css
manifest/manifest.json
requirements/Requirements_Matrix.md
traceability/Traceability_Matrix.md
journal/Engineering_Journal.md
diagrams/Architecture.svg
diagrams/Component.svg
diagrams/Workflow.svg
diagrams/EventFlow.svg
diagrams/Integration.svg
examples/example_threat_intelligence.md
```

58. Final Archive Statement

EA-0014 is the Engineering Archive for IS-014 - AQELYN Threat Intelligence Fusion Engine.

It records the completed specification, the architectural decisions, the integration model, the repository impact, the risk posture, verification requirements, acceptance criteria, archive index update, and the engineering publication standard.

```
EA-0014 COMPLETE
IS-014 COMPLETE
NEXT: IS-015
```